

Jack Cutrara

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PROJECTS

SeatWatch | Live Reservation Booking SaaS | seat.watch

2025 - Present

TypeScript, React 19, Express, PostgreSQL, Redis, BullMQ

- Architected and shipped a production SaaS platform across three independent services (React 19 SPA, Express REST API, BullMQ worker) deployed on Railway with PostgreSQL 16 and Redis 7, totaling 56,000+ lines of TypeScript across 375 source files with comprehensive Vitest and Playwright test coverage.
- Built a dual-strategy booking engine that executes via direct API integration in sub-second time, with automatic fallback to a Patchright headless browser session on upstream API changes, preserving uptime through provider-side modifications.
- Engineered horizontally-scalable concurrency primitives in Redis: per-user circuit breakers (CLOSED, OPEN, HALF_OPEN state machine), distributed booking locks via SET NX EX with Lua-script atomic release, and a token-bucket global rate limiter. Together they enable safe multi-worker deployment without double-booking.
- Designed a per-user execution isolation layer with 50+ deterministically-assigned browser identity profiles, Impit-powered TLS fingerprinting for browser-grade handshakes on direct API calls, and Poisson-distributed poll scheduling to maintain stable monitoring across concurrent users.
- Delivered end-to-end Stripe billing across four pricing tiers with transaction-level plan enforcement, plus Clerk-authenticated JWT auth, AES-256-GCM credential encryption at rest, 13 React Email transactional templates, and Sentry-monitored production deployment.

Multi-Chain EVM Trading System | github.com/jackc625

2025 - Present

TypeScript, Node.js, Solidity, ethers v6, SQLite, React

- Built a multi-chain trading runtime in TypeScript on Node 22 spanning Ethereum, BSC, Base, and Arbitrum. Abstracted chain-specific configuration behind discriminated TypeScript unions with exhaustive never-checks that force-compile any unhandled chain or DEX variant.
- Implemented three MEV-protected transaction strategies (Flashbots bundles, bloXroute private transactions, EIP-1559 direct submission) behind a single MevStrategy interface, dispatched per-chain with strategy-specific retry and health-check protocols.
- Designed a unified exit state machine with per-token sell mutex, volatility-adjusted trailing stops, age-weighted urgency multipliers, and full state persistence to SQLite. Survives SIGKILL and resumes every position at its persisted high-water mark on restart.
- Authored and shipped an immutable, stateless, eth_call-only Solidity contract for pre-trade safety verification. Audited with Slither and Mythril under an evergreen CI gate, deployed via Hardhat Ignition with a capability-split deploy key isolated from the runtime wallet.

NFL Prediction System | Full-Stack ML Pipeline | github.com/jackc625

2025 - Present

Python, FastAPI, XGBoost, DuckDB, scikit-learn, HTMX

- Designed a three-layer data pipeline (Bronze/Silver/Gold) ingesting play-by-play, market odds, and weather data. Engineered 55-70 features per game with Pydantic validation at every boundary and DuckDB + Parquet dual storage.
- Trained three specialized models: logistic regression with isotonic calibration for Win Probability, and XGBoost regressors for Against-the-Spread and Over/Under. All trained under a walk-forward temporal validation harness that hard-fails on any future-data contamination.
- Built a LeakageGate validator that enforces time-fence cutoffs at the FeatureBuilder layer and scans the final feature matrix for leakage keywords, making no-leakage a compile-time invariant rather than a convention.
- Shipped a FastAPI + Jinja2 + HTMX dashboard with Plotly visualizations, walk-forward backtest reporting across the 2021-2024 seasons, and parallel betting simulators (flat-stake and quarter-Kelly), automated via a scheduled GitHub Actions workflow timed to the Friday odds snapshot.

PROFESSIONAL EXPERIENCE

Project Management Intern | Balfour Beatty, Chantilly, VA

May 2023 - Aug 2023

- Tracked deliverables and subcontractor timelines across active construction workstreams; coordinated with engineers, subcontractors, and clients to keep multi-phase milestones on schedule.
- Led stakeholder meetings and prepared status reports for project leadership, surfacing risks and blockers in real time.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, SQL, Solidity, HTML, CSS

Frontend: React, Next.js, Preact, Tailwind CSS, shadcn/ui, Zustand, TanStack React Query, HTMX

Backend & Data: Node.js, Express, FastAPI, Spring Boot, BullMQ, PostgreSQL, Redis, SQLite, DuckDB, Prisma, Supabase, Pydantic, Zod

Tools & DevOps: Docker, Railway, AWS S3, GitHub Actions, Git, Hardhat, Playwright, Vitest, pytest, Stripe, Clerk, Sentry

EDUCATION

Bachelor of Science, Computer Science | Western Governors University

May 2026

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Databases, Software Engineering

Previously: Virginia Tech (2020-2023). Transferred to complete CS degree.

Certifications: LPI Linux Essentials (2025)